

Barriers to Entry and Regional Economic Growth in China

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Motivation

- Since the onset of economic reform in the late 1970s, China has gone from one of the poorest to a middle-income economy
- Expansion of non-state sector was main source of growth (Zhu, 2012)
- But growth was highly uneven across localities (≈ 350 prefectures)
- We show that
 - : By mid-1990s, there were **sizable local differences** in productivity, wages, & size of non-state manufacturing sector
 - dispersion reflected divergence before 1995
 - : Reversal of fortune from mid-1990s: differences across localities in non-state manufacturing performance started disappearing
 - **strong convergence across prefectures** in non-state value added per worker, TFP, wages, and capital per worker

Overview

- Aim of paper: understand forces behind initial dispersion and 1995-2008 convergence
- Brandt, van Biesebroeck, and Zhang (2012) argue: creation and selection of new firms is most important source of non-state sector productivity and output growth
- We find: this process is very different across prefectures:
 - ... in prefectures with a large presence of state firms,
 - : **less entry** of non-state firms
 - : non-state entrants pay **lower wages**, have **lower TFP**,
lower value added per worker, **lower capital per worker**

Overview

- Build closed economy version of Hopenhayn (1992) model with 3 distortions to account for empirical patterns
 - : **capital** and **output** wedges,
 - : an **entry** wedge
- Interpretation of entry wedge: restriction on number of licences allowing potential entrants to operate
- Solve model analytically
- Estimate model using firm-level data from the 1995, 2004, and 2008 Chinese Industrial Census
- Develop a political economy model to rationalize local government behavior

Findings: Entry Wedge Is Quantitatively Most Important

- Entry wedge:
 - : main driver of initial 1995 dispersion
 - : main driver of 1995-2008 convergence
- World Bank survey “Cost of Doing Business in China, 2008”
 - : indices match well with our 2008 entry wedge estimates
- Study the empirical factors behind measured entry wedges:
 - : 1995 level systematically linked to size of SOE sector
 - : convergence after 1995 tied to downsizing of state sector

Chinese Industrial Census

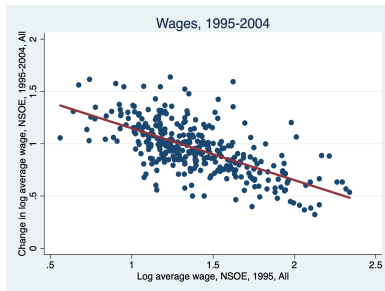
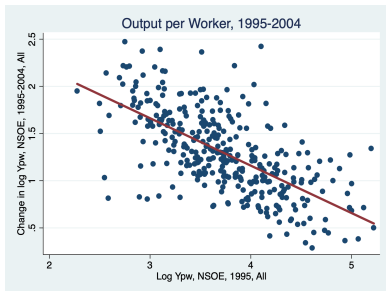
- Chinese Industrial Census (CIC)
- CIC: (1992), 1995, 2004, 2008
- Large: covers most of the manufacturing sector
- Firm-level records on 0.53 (1995), 1.37 (2004), and 2.08 (2008) million firms
- Rich: firm-level observations on value added, employment, capital stock, wage bill, year of birth, ownership, sector
- Data work (issues)
 - make prefectures consistent across years
 - define the SOE sector (especially in 2004 and 2008)
 - construct measures of real capital

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Regional Dispersion and Convergence

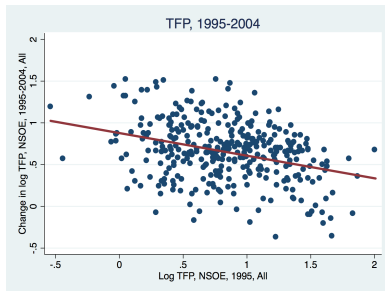
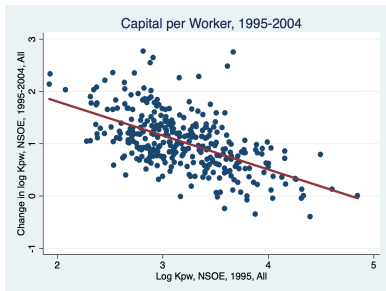
	$\frac{Y}{N}$	w	$\frac{K}{N}$	Aggr. TFP Z
β -convergence 1995-2004	7.8%	7.6%	11.7%	3.7%
β -convergence 1995-2008	9.2%	5.7%	11.9%	5.1%
90-10 ratio in 1995	4.4	2.3	3.5	3.1
90-10 ratio in 2004	2.8	1.8	3.1	3.2
90-10 ratio in 2008	3.1	3.0	4.2	2.9

Non-State Dispersion & Convergence, 1995-2004



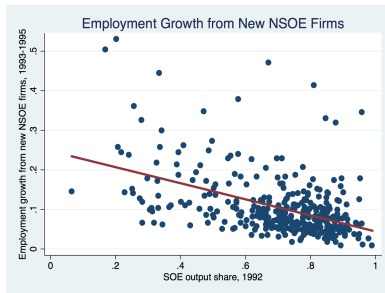
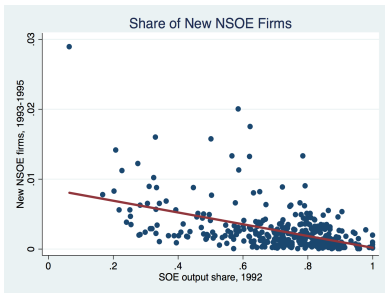
- 1995 (90/10) ratio: Ypw 4.4; wages 2.3
- rate of convergence, **output per worker**; **7.8%** after 1995
(Barro and Sala-i-Martin: regional convergence rate in USA: 2%)
- rate of convergence, **wages**; **7.6%** after 1995

Non-State Dispersion & Convergence, 1995-2004

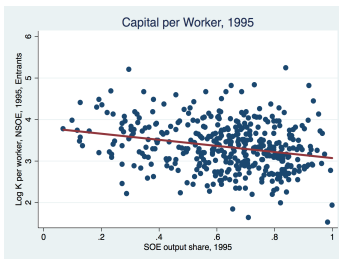
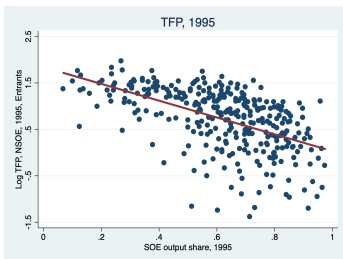
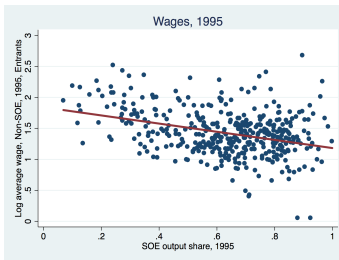
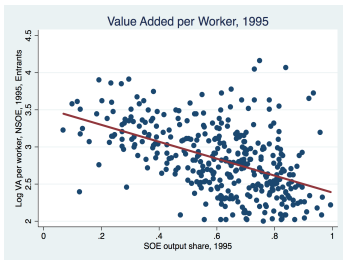


- 1995 (90/10) ratio: Kpw 3.5; TFP 3.1
- rate of convergence, **capital per worker**; **11.7%** after 1995
- rate of convergence, **TFP** (calculated as Solow residual); **3.7%** after 1995

Size of State Sector and NSOEs in 1995: Firm Entry



Size of State Sector and NSOEs in 1995: VA/N, Wages, TFP, K/N



Version of a Hopenhayn (1992) Model

$$y_i = z_i^{1-\eta} \left(k_i^{1-\alpha_j} n_i^{\alpha_j} \right)^\eta$$

- firms in each industry have common production function
- $j = J(i)$ denotes industry for firm i
- $0 < \eta < 1$: decreasing returns to scale
- common rental rate of capital ($r + \delta$)
- closed labor market: prefecture-specific wage rate w
- distortions: output tax τ_i^Y and capital tax τ_i^K
- Benchmark: focus on prefecture-specific wedges.
 - : Extension: allow within-prefecture firm heterogeneity

Firm's Problem: Output and Capital Wedges

- The firm's objective is

$$\max_{k,n} \left\{ (1 - \tau^y) y - wn - (1 + \tau^k) (r + \delta) k \right\}.$$

- Firms' FOCs for k and n imply linear allocations in z

$$y^* = z \left((1 - \tau^y) \eta \right)^{\frac{\eta}{1-\eta}} \left(\frac{1 - \alpha}{(1 + \tau^k) (r + \delta)} \right)^{\frac{(1-\alpha)\eta}{1-\eta}} \left(\frac{\alpha}{w} \right)^{\frac{\alpha\eta}{1-\eta}}$$

$$\equiv Z \cdot z \cdot \bar{y}$$

$$n^* = n(z, \tau^y, \tau^k; w) = Z \cdot z \cdot \alpha \eta \left(\frac{1 - \tau^y}{w} \right) \cdot \bar{y}$$

$$k^* = k(z, \tau^y, \tau^k; w) = Z \cdot z \cdot (1 - \alpha) \eta \frac{1 - \tau^y}{(1 + \tau^k) (r + \delta)} \cdot \bar{y}$$

$$\Pi^* = Z \cdot z \cdot (1 - \tau^y) (1 - \eta) \cdot \bar{y}.$$

Entrepreneur's Problem

- Large (but finite) number M of potential entrepreneurs in each prefecture
- Potential entrepreneurs observe individual TFP z
- z is Pareto distributed $f(z) = \underline{z}^{\xi} z^{-\xi-1}$
: where $\xi > 1$, $\underline{z} \geq 1$, and $z \in [\underline{z}^{1/\xi}, \infty)$
- Entrepreneur incurs fixed cost v if firm is operated

Entry Decision

- Must recover fixed cost to find it optimal to operate: $\Pi(z^*) > v$.
- Threshold property: only entrepreneurs with $z \geq z^*$ will operate, where

$$z^*(\tau^y, \tau^k; w) = \frac{v}{Z(1 - \tau^y)(1 - \eta) \cdot \bar{y}}$$

Equilibrium: Clearing of the Labor Market (with entry barrier)

- Entry barrier: only a share $(1 - \psi)$ of potential entrants allowed to enter
 - : random selection/lottery
- Take the wedges and (N_j, M_j) as exogenous
 - : abstract from endogenous migration decisions of workers and entrepreneurs
 - : realized empirical distribution of (N_j, M_j) affected by worker and entrepreneur migration
- Equilibrium wage w clears the (local) labor market

$$M(1 - \psi) \int_{z^*}^{\infty} n(z, \tau^y, \tau^k; w) f(z) dz = N$$

Equilibrium Wage: w

$$\begin{aligned}\ln w &= \mu(1-\eta) \ln \left[Z^\xi (1-\psi) \frac{M}{N} \right] \\ &\quad + \mu \xi \ln (1-\tau_y) \\ &\quad - \mu(1-\alpha) \xi \eta \ln [(1+\tau_k)(r+\delta)] \\ &\quad + \Omega(\alpha, \eta, \xi, \nu)\end{aligned}$$

- $\mu \equiv \frac{1}{1-\eta+\xi\alpha\eta} > 0$
- $\Omega(\alpha, \eta, \xi)$ is a constant

Equilibrium: Solow Residual S

$$\begin{aligned}\ln S = & \mu \alpha \eta (1 - \eta) \ln \left[Z^{\xi} (1 - \psi) \frac{M}{N} \right] \\ & - \mu (1 - \eta) \ln (1 - \tau_y) \\ & + \mu (1 - \eta) [1 + (\xi - 1) \alpha \eta] \ln [(1 + \tau_k)(r + \delta)] \\ & + \hat{\Omega}(\alpha, \eta, \xi, \nu)\end{aligned}$$

- Solow residual S defined as $Y = SK^{1-\theta}N^{\theta}$, given a labor share θ
- $\mu \equiv \frac{1}{1-\eta+\xi\alpha\eta} > 0$
- $\hat{\Omega}(\alpha, \eta, \xi, \nu)$ is a constant

Effect of Wedges and N/M

	$(1 - \tau_y)$	$(1 + \tau_k)$	$(1 - \psi)$	N/M
wage rate w	+	-	+	-
Solow residual S	-	+	+	-
Entry Γ	+	-	+	+

[More]

Equilibrium Mechanism

- Suppose $(1 - \psi)$ is small
- Low $(1 - \psi)$ implies that few firms enter
- Low entry implies low wages required to clear the labor market (since little competition for workers)
- Low wages implies low z^* (since labor is cheap)
- Low z^* implies negative selection
... hence low TFP and low Y/N

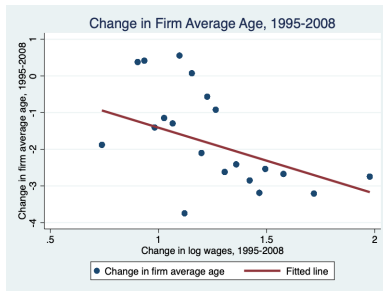
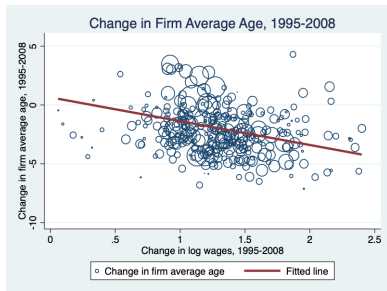
Evidence for Selection Mechanism

Dynamic extension of theory

- Aggregate TFP shifts should affect labor demand for all firms the same. Should have no effect on firm entry
- A lower entry barrier should increase entry and labor demand for entrants, no effect for incumbent firms
- Theory predicts that
 - If wage growth is driven by changes in entry barrier: wage growth and firm age should be negatively correlated
 - If wage growth is driven by aggregate TFP shifts: wage growth should be uncorrelated with firm age

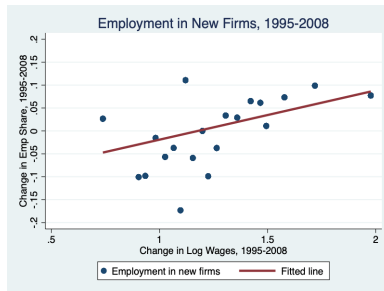
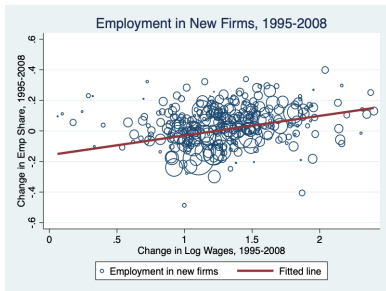
Evidence for Selection Mechanism

Figure: Change in Firm Average Age, by Prefecture, 1995-2008

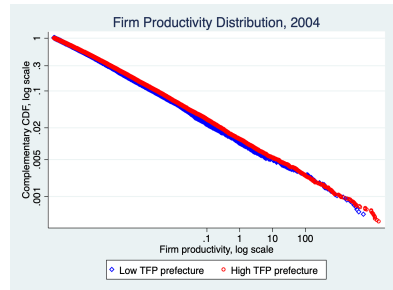
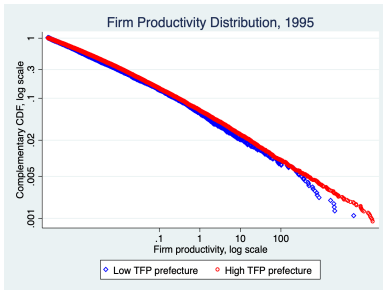


Evidence for Selection Mechanism

Figure: Employment Share in New Firms, by Prefecture, 1995-2008



Evidence for Selection Mechanism: Upper Tail of z Distribution



: distribution if z is above the 95th percentile in the overall distribution

Measuring the Wedges

- Labor share for each industry $\alpha\eta$: Hsieh and Klenow (2009)
- Decreasing returns: $\eta = 0.85$ (Restuccia and Rogerson 2008)
- $\xi = 1.05$, Pareto parameter, use 30% of the most productive firms

$$\frac{E(z|z \geq z^*)}{z^*} = \frac{\xi}{\xi - 1}$$

- Set v such that $n^*(z^*) = 1$ in the lowest s prefectures
- Set $\underline{z}$ such that $\psi = 0$ in the lowest s prefectures
- From 1995, 204, 2008 Chinese Industrial Census
 - value added: y_i
 - wage bill: $w_i n_i$
 - estimated real capital: k_i

Output and Capital Wedges

- τ_i^y and τ_i^k identified from firm's first-order conditions, for k and n

$$(1 - \tau_i^y) = \frac{1}{\alpha_j \eta} \frac{w_i n_i}{y_i}$$

$$(1 + \tau_i^k) = \frac{1 - \alpha_j}{\alpha_j} \frac{w_i n_i}{(r + \delta) k_i}$$

- Gross output wedge in the prefecture, Δ_p^y

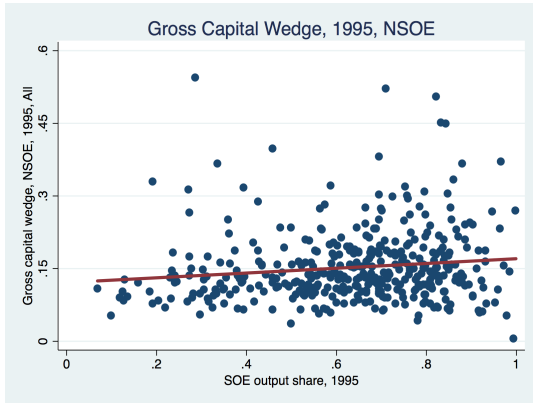
$$\Delta_p^y = \sum_{j=1}^J \left(\frac{1}{\alpha_j \eta} \sum_{i \in (j,p)} \frac{w_i n_i}{y_i} \frac{y_i}{Y_{j,p}} \right) \frac{Y_{j,p}}{Y_p},$$

- Gross capital wedge in the prefecture, Δ_p^k

$$\Delta_p^k = \sum_{j=1}^J \left(\frac{1 - \alpha_j}{\alpha_j} \sum_{i \in (j,p)} \frac{w_i n_i}{k_i} \frac{k_i}{K_{j,p}} \right) \frac{K_{j,p}}{K_p}.$$

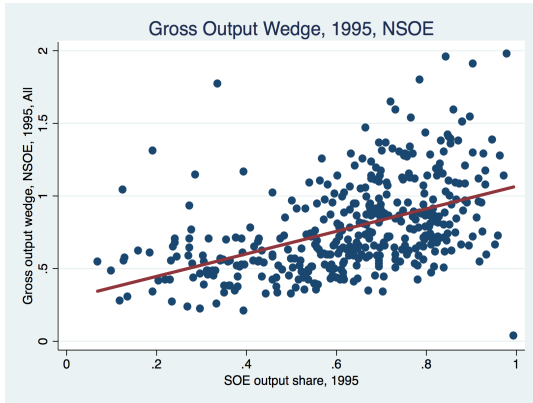
- Compute Δ_p^y and Δ_p^k for each prefecture in the dataset

Gross Capital Wedge: Δ_p^k



- Capital taxes slightly higher in high SOE-share prefectures

Gross Output Wedge: Δ_p^y



- Output taxes low in high SOE-share prefectures

Entry Wedge ($1 - \psi_p$)

Estimate ψ_p in prefecture p from the equilibrium wage equation

$$\begin{aligned}\ln(1 - \psi_p) = & \frac{1 - \eta + \xi \alpha \eta}{1 - \eta} \ln w_p \\ & - \frac{\xi}{1 - \eta} \ln \Delta_p^y \\ & + \frac{\xi \eta (1 - \alpha)}{1 - \eta} \ln \Delta_p^k \\ & + \ln(N_p/M_p) \\ & + \bar{\Omega}(\alpha(p), \eta, \xi, \underline{z}, v)\end{aligned}$$

- $\bar{\Omega}(\alpha, \eta, \xi, v)$ is a constant

Measuring the (N/M) Ratio: Migration of Workers and Entrepreneurs

	(1) $m_n \equiv \frac{\text{migrant workers}}{\text{total workers}}$	(2) $m_e \equiv \frac{\text{migrant entrepreneurs}}{\text{total entrepreneurs}}$	(3) $\text{corr}(m_n, m_e)$
1990	0.055	0.024	0.527
2000	0.249	0.115	0.346
2010	0.304	0.220	0.340

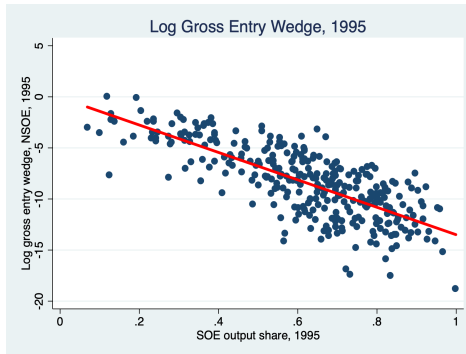
- Chinese Population Census: 1990, 2000, and 2010
 - identify migrants: live and work in a location other than the one in which they have their Hukou
 - we use Chief Executive Officer (CEO) as a proxy for entrepreneur: owner of a firm is in most cases also the CEO, and vast majority of firms in China are controlled by an individual with more than 50% of the shares
- workers and entrepreneurs migration rates positively correlated
- migration rates are also positively correlated with wages

Measuring the (N/M) Ratio

$$\frac{N}{M} = \underbrace{\left(1 - \frac{N_{SOE}}{N_D + N_I}\right)}_{\text{NSOE share}} \underbrace{\frac{1 - m_e}{1 - m_n}}_{\text{non-migrant ratio}}$$

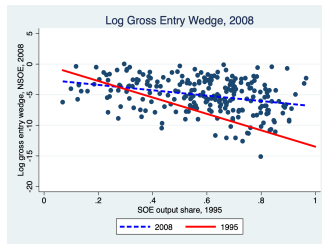
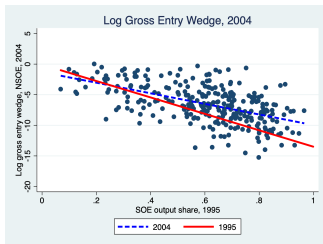
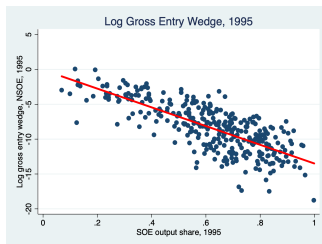
- m_e : entrepreneur migration rate
- m_n : worker migration rate
- N_{SOE} : fraction of workers employed by non-state (NSOE) firms
- N_D : local manufacturing workers
- N_I : migrant manufacturing workers

Estimated NSOE Entry Wedge ($1 - \psi_p$) in 1995



- Log gross entry wedge $\ln(1 - \hat{\psi}_p)$
- SOE share accounts for 52% of the variation in the entry wedge

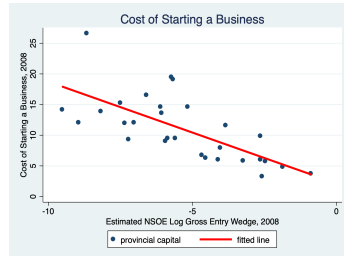
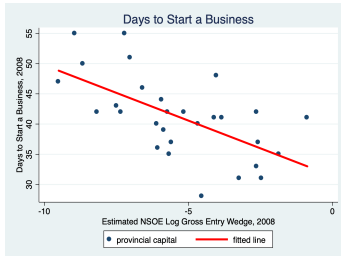
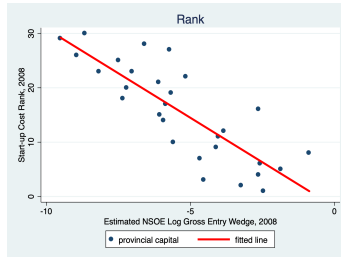
Estimated NSOE Entry Wedge ($1 - \psi_p$)



2008 Costs of Starting a Business in China

- “Doing Business in China 2008” Report
 - : The World Bank Group (2008)
 - : provides various measures of the cost of starting a business in main provincial cities
- Measures
 - : Rank: from easy (1) to hard (30) to start a business
 - : Days it takes to start a business
 - : Cost of starting a business: as a % of provincial GDP per capita

“Doing Business in China” and Entry Wedges, 2008



Entry Rates and Wedges

- Non-SOE entry rates were not targeted in the estimation of the model
- Entry rate measure $\Gamma_{p,t}^e$ for prefecture p in period $t = 1995, 2004, 2008$

$$\Gamma_{p,t}^e = \frac{N_{p,t}^e}{N_{p,t} - N_{p,t}^e}$$

- : $N_{p,t}^e$ is employment in new non-SOE firms
- : $N_{p,t}$ is total employment
- : new firms are started in period $t - 1$ or $t - 2$
- : firms started in period t are dropped

Entry Rates and Wedges

$$\ln \Gamma_{p,t}^e = \beta_0 + \underbrace{\beta_1 \ln(1 - \tau_{p,t}^y)}_{(+)\text{ log gross output wedge}} + \underbrace{\beta_2 \ln[(1 + \tau_{p,t}^k)(r + \delta)]}_{(-)\text{ log gross capital wedge}} + \underbrace{\beta_3 \ln(1 - \psi_{p,t})}_{(+)\text{ log gross entry wedge}} + \varepsilon_{p,t}$$

	$\ln(1 - \tau^y)$		$\ln(1 + \tau^k)$		$\ln(1 - \psi)$	
	β_1	1sd	β_2	1sd	β_3	1sd
1995	0.256*	10.6%	-0.216**	-9.2%	0.098***	33.9%
2004	0.101	3.4%	-0.052	-1.9%	0.032**	10.1%
2008	0.143	6.74%	-0.127	-6.8%	0.013	3.6 %

Note: ** – statistically significant at 1%; * – statistically significant at 10%.

Entry Rates and Wedges

$$\Delta \ln \Gamma_{p,t}^e = \gamma_0 + \underbrace{\gamma_1 \Delta \ln(1 - \tau_{p,t}^y)}_{(+)} + \underbrace{\gamma_2 \Delta \ln[(1 + \tau_{p,t}^k)(r + \delta)]}_{(-)} + \underbrace{\gamma_3 \Delta \ln(1 - \psi_{p,t})}_{(+)} + \varepsilon_{p,t}$$

log gross output wedge log gross capital wedge log gross entry wedge

	$\Delta \ln(1 - \tau^y)$		$\Delta \ln(1 + \tau^k)$		$\Delta \ln(1 - \psi)$	
	γ_1	1sd	γ_2	1sd	γ_3	1sd
1995-2004	0.689***	23.0%	-0.508***	-21.6%	0.116 ***	28.7%
1995-2008	0.504***	26.7%	-0.569***	-32.8%	0.103 ***	31.2%

Note: ** – statistically significant at 1%; * – statistically significant at 10%.

Convergence in TFP and Wages

		TFP		Wages	
Change in		1995-2004	1995-2008	1995-2004	1995-2008
all		0.041	0.059	0.081	0.066
$(1 - \psi)$		0.042	0.050	0.032	0.044
$(1 + \tau_k)$		-0.005	-0.003	0.014	0.019
$(1 - \tau_y)$		0.011	0.015	-0.004	-0.020
$\alpha\eta$		-0.004	-0.004	0.025	0.024
N/M		-0.001	-0.001	0.003	-0.001
-	m_N	0.002	0.002	0.011	0.009
-	m_e	-0.001	-0.001	-0.002	-0.004
-	NSOE share	-0.003	-0.002	-0.005	-0.005

What Explains the Entry Wedges?

$$\ln(1 - \psi)_{p,t} = \beta_0 + \beta e_{p,t}^{SOE} + X_{p,t}\gamma' + \varepsilon_{p,t}$$

$$\Delta \ln(1 - \psi)_{it} = \beta_0 + \beta_1 \Delta e_{it}^{SOE} + \Delta X_{it}\gamma' + \Delta \varepsilon_{it}$$

Controls

- $\ln FREV_t$: 1995 (2004) log fiscal revenue per government worker
- $\ln PROF_t^{soe}$: 1995 ratio of profits to total assets for SOEs
- $e_p^{soe} = \frac{E_p^{soe}}{E_p}$: 1995 (2004, 2008) share of SOE employment in pref. p

Cross-Sectional IVs for SOE Share, e_p^{soe}

Cross-sectional instruments: **lagged variables**

- $IV_{lag} = e_{p,t-1}^{soe}$; lagged SOE employment share in pref. p
- IV_{1978}
 - : restrict 1995 sample to firms established 1978 or earlier
 - : measure SOE share in 1978 using this restricted sample
- IV_{prov} : use 1978 GDP provincial data and construct province SOE share in 1978

The Entry Wedge in 1995, 2004, and 2008

	$\ln(1 - \psi)$	<i>OLS</i>	IV_{lag}	IV_{1978}	IV_{prov}
1995	e^{SOE}	-12.42***	-15.36***	-13.63***	-14.44***
	$\ln FREV$	1.32***	0.88*	1.14**	0.73
	$\ln PROF^{SOE}$	0.24*	0.24*	0.24*	0.19
2004	e^{SOE}	-8.61***	-12.30***	-14.44***	-18.75***
	$\ln FREV$	1.48***	1.08**	0.84*	0.80
2008	e^{SOE}	-6.79***	-7.97***	-8.35***	-4.85

Note: *** – statistically significant at 1%; ** – at 5%; * – at 10%.

[First-Stage Results]

Time-Series IV for Change in SOE Share, Δe_p^{soe}

- **Bartik** instrument for 1995-2004 SOE empl. change

: 1998 SOE reform “Grab the Large, Release the Small”

- Aggregate 1995-2004 SOE empl. change in industry j

$$: \mu_j^{soe} = \frac{E_{j,2004}^{soe} - E_{j,1995}^{soe}}{E_{j,1995}^{soe}}$$

- 1995 ratio SOE empl. share in ind. j / pref. p empl.

$$: e_{p,j}^{soe} = \frac{E_{p,j}^{soe}}{E_p}$$

- Predicted increase in SOE employment (**Bartik** instrument)

$$: IV_p^{ind} = \sum_j e_{p,j}^{soe} * \mu_j^{soe}$$

Change in the Entry Wedge, 1995-2004

$\Delta \ln(1 - \psi)$	<i>OLS</i>	<i>OLS</i>	<i>IV_{Bartik}</i>	<i>IV_{Bartik}</i>
ΔS	-2.06* (1.16)	-1.82 (1.16)	-8.99** (2.63)	-10.28*** (2.91)
$\Delta \ln FREV$		1.24*** (0.40)		0.76* (0.46)
<i>First stage:</i>				
<i>IV coefficient</i>			0.62***	0.65***
<i>st. error</i>			(0.07)	(0.08)
R^2			0.22	0.25

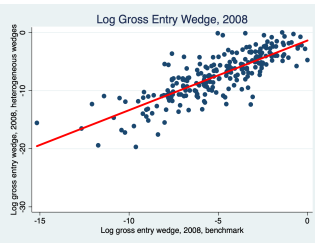
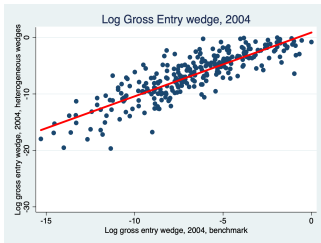
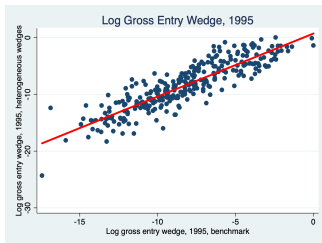
Note: *** – statistically significant at 1%; ** – at 5%; * – at 10%.

State Sector and Regional Convergence, 1995-2004

	$\Delta \ln w$	$\Delta \ln(VA/N)$	$\Delta \ln TFP$	$\Delta \ln(K/N)$
$\Delta \hat{S}_p$	-0.61*** (0.18)	-1.24*** (0.30)	-0.33 (0.22)	-1.90*** (0.40)
$\ln x_{1995}$	-0.36*** (0.05)	-0.47*** (0.05)	-0.34*** (0.04)	-0.51*** (0.07)
Province F.E.	Yes	Yes	Yes	Yes
First stage				
β_{IV}	0.83	0.85	0.87	0.86
s.e.	(0.09)	(0.09)	(0.09)	(0.09)
adj. R^2	0.38	0.36	0.36	0.36

Note: *** – statistically significant at 1%; ** – at 5%; * – at 10%.

Extension: Heterogeneity of Wedges across Firms

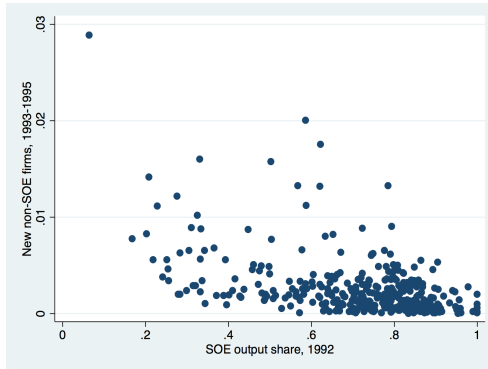


Conclusion

- Study growth patterns of non-state sector across localities in China
- Build Hopenhayn model of new firm entry with multiple distortions
- Identify novel entry wedge as key to explaining heterogeneity in new firm behavior across prefectures
 - Provide out-of-sample validation for these wedges
 - Link size and changes of entry costs to dynamics of state-sector
- Develop political-ec. model of local government behavior to motivate observed correlations between entry wedges and SOE presence
- Future directions
 - Allow wedges to differ by industry and location
 - Extend through Great Recession to capture possible reversal
 - Study role of wedges for impeding structural transformation

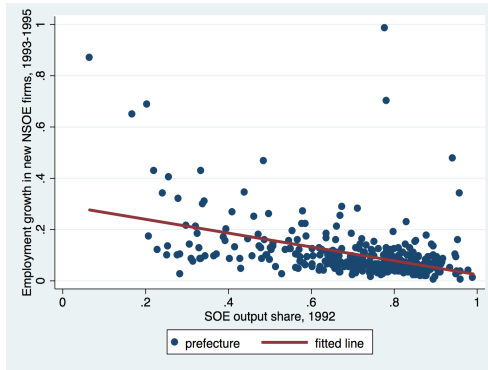
Additional Slides

Firm Entry in the Non-state Sector, 1995



- Distribution of new non-state firms (1993-1995 entrants)
- Most are in the low s prefectures

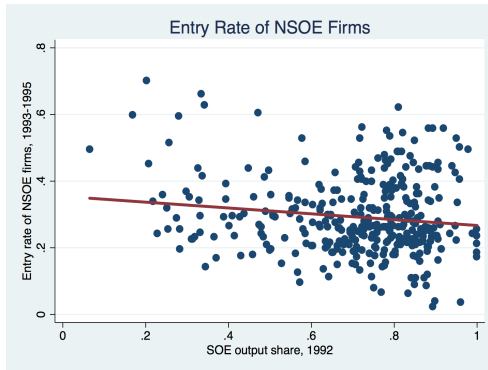
Firm Entry in the Non-state Sector, 1995



- Employment in new non-state entrants (1993-1995) relative to the employment in all firms in 1992
- Lower in high s prefectures

[Number of firms]

Firm Entry in the Non-state Sector, 1995



- New non-state entrants (1993-1995) relative to the stock of all firms in 1992
- Lower in high s prefectures

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Framework for Wedges: The Labor Wedge

- Incorporating the gross labor wedge: $(1 + \tau^w)$
- Gross output wedge, Δ_i^y

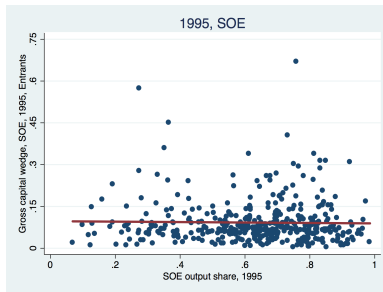
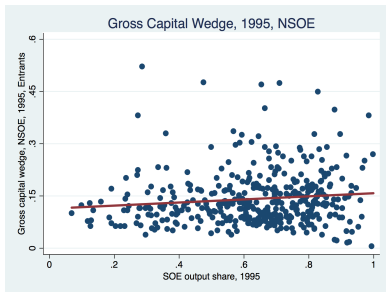
$$\Delta_i^y = \frac{(1 - \tau_i^y)}{(1 + \tau^w)} = \frac{1}{\alpha \eta} \frac{w_i n_i}{y_i}$$

- Gross capital wedge, Δ_i^k

$$\Delta_i^k = \frac{(1 + \tau_i^k)(r + \delta)}{(1 + \tau^w)} = \frac{1 - \alpha}{\alpha} \cdot \frac{w_i n_i}{k_i}$$

- If the labor wedge increases with s , then in the NSOE sectors
 - : the output subsidies have to be even higher in the high s prefectures, and
 - : the capital tax wedges have to be higher in the high s prefectures

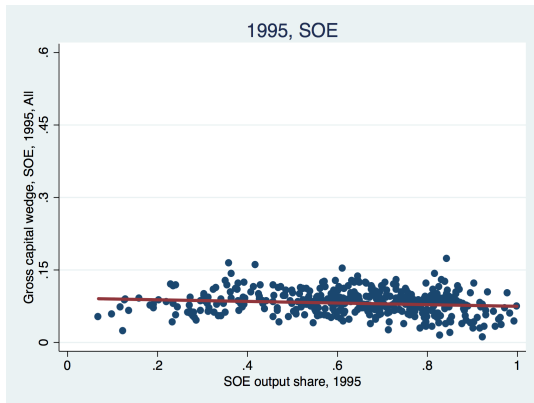
Gross Capital Wedge, Entrants: Δ^k



- Higher capital taxes in high s prefectures for non-SOE firms
- No relationship between capital taxes and s for SOE firms

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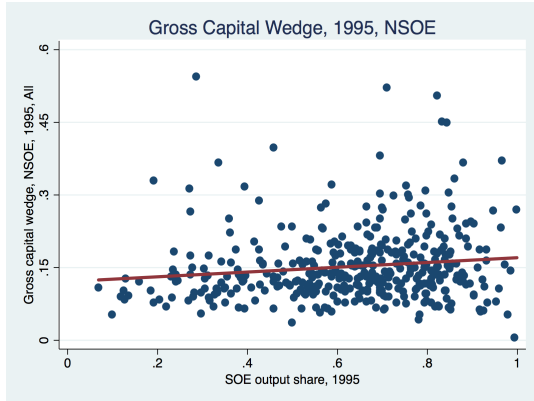
Gross Capital Wedge: Δ^k



- No relationship between capital taxes and s for SOE firms

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Gross Capital Wedge: Δ^k

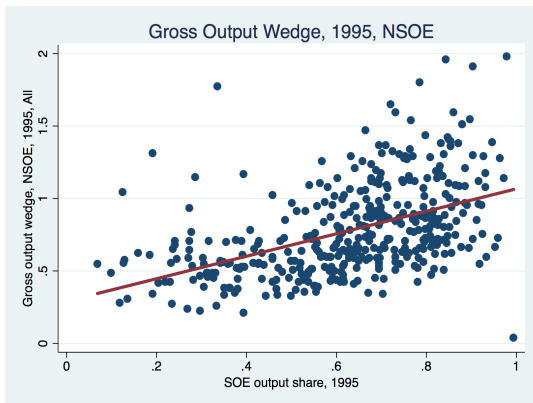


- Higher capital taxes in high s pref. for non-SOE firms

[Entrants]

[SOEs]

Gross Output Wedge: Δ^y

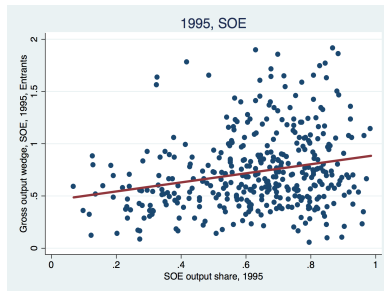
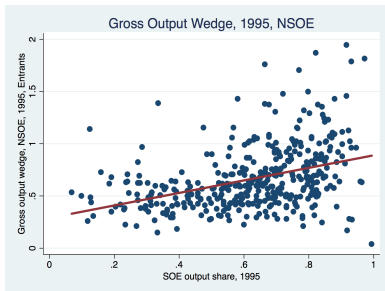


- Lower output taxes (higher subsidies) in high s pref. for non-SOE firms

[Entrants]

[SOEs]

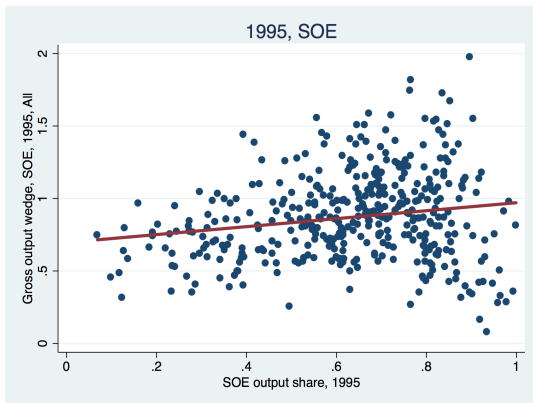
Gross Output Wedge, Entrants: Δ^y



- Lower output taxes (higher subsidies) in high s prefectures
- For both non-SOE and SOE firms

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Gross Output Wedge: Δ^y



- Lower output taxes (higher subsidies) in high s pref. for SOE firms

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Entry Decision

- $f(z)$ is Pareto distributed

$$f(z) = \underline{z}^{\xi} \xi z^{-\xi-1},$$

$$\begin{aligned} &: \xi > 1 \\ &: \underline{z} \geq 1, z \in [\underline{z}, \infty) \end{aligned}$$

- The firm problem implies:

$$\begin{aligned} y &= z((1 - \tau^y)\eta)^{\frac{\eta}{1-\eta}} \left(\frac{1 - \alpha}{(1 + \tau^k)(r + \delta)} \right)^{\frac{(1-\alpha)\eta}{1-\eta}} \left(\frac{\alpha}{w} \right)^{\frac{\alpha\eta}{1-\eta}} \\ &\equiv z \cdot \bar{y} \\ n &= z \cdot \alpha \eta \left(\frac{1 - \tau^y}{w} \right) \cdot \bar{y} \\ k &= z \cdot (1 - \alpha) \eta \frac{1 - \tau^y}{(1 + \tau^k)(r + \delta)} \cdot \bar{y} \\ \Pi &= z \cdot (1 - \tau^y)(1 - \eta) \cdot \bar{y}. \end{aligned}$$

Entry Decision

- Only entrepreneurs with $z \geq z^*$ will operate, where

$$z^* = \frac{v}{(1 - \tau^y)(1 - \eta) \cdot \bar{y}}$$

- The measure Γ of all operating entrepreneurs is

$$\Gamma(z \geq z^*) = M(1 - \psi) \int_{z^*}^{\infty} \underline{z}^{\xi} \xi z^{-\xi-1} dz = M(1 - \psi) \underline{z}^{\xi} (z^*)^{-\xi}$$

- The equilibrium wage w clears the labor market

$$M(1 - \psi) \int_{z^*}^{\infty} n(z) f(z) dz = N$$

- Normalize by the size of the labor force in the prefecture

The Effect of the Wedges

	$(1 - \tau^Y)$	$(1 + \tau^k)$	$(1 - \psi)$
w	$\mu\xi > 0$	$-\mu(1 - \alpha)\xi\eta < 0$	$\mu(1 - \eta) > 0$
TFP_s	$-\mu(1 - \eta) < 0$	$\mu(1 - \eta)[1 + (\xi - 1)\alpha\eta] > 0$	$\mu\alpha\eta(1 - \eta) > 0$
Entry	$\mu\xi > 0$	$-\mu\xi(1 - \alpha) < 0$	$\mu(1 - \eta) > 0$
$\frac{Y}{N}$	$\mu\xi\eta(1 - \alpha) +$ $\mu(\xi - 1)(1 - \eta) > 0$	$-\mu(1 - \alpha)\xi\eta < 0$	$\mu(1 - \eta) > 0$

- $\mu = \frac{1}{1 - \eta + \xi\alpha\eta} > 0$

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Understanding the Entry Wedge

- 1995, the entry wedge is higher in prefectures where
 - : the share of employment (or output) in the SOE sector is higher
 - : fiscal revenues per government worker are lower
 - : the profitability of SOEs is lower
- 1995-2004, the decline in the entry wedge is larger in pref. where
 - : the decline in the SOE share of employment is larger
 - : the increase in fiscal revenues per government worker are larger

Note that data on

- : fiscal revenue per government worker available for 1995 and 2004
- : profitability of SOEs available for 1995

Fiscal and SOE Reforms

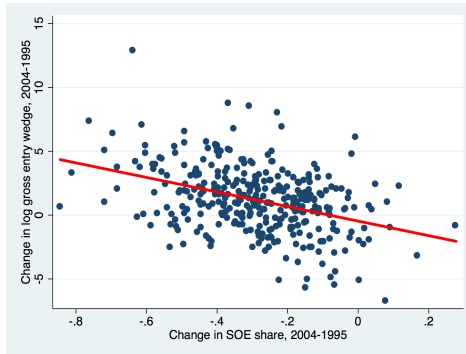
- SOE reforms after 1995
 - : smaller SOEs sold off or shutdown
 - : massive layoffs of workers in the SOE sector including in those firms not privatized
 - : concentration of SOEs in strategic and pillar sectors
- Fiscal reform after 1995
 - : recentralization of the fiscal system that increased the % of revenue going to the center
 - : new system of fiscal transfers and sharing rules between provinces and the center, and localities and provinces
 - : localities allowed to retain land conveyance fees; i.e., basically profits from the sale of farm land for non-agricultural uses

The Entry Wedge in 1995, 2004, and 2008

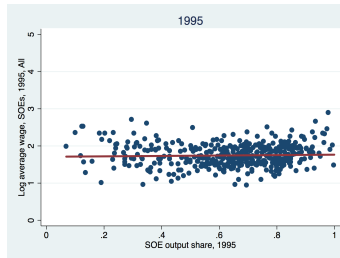
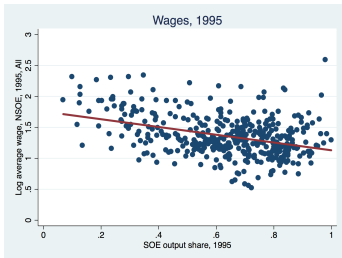
	$\ln(1 - \psi)$	OLS	IV_{lag}	IV_{1978}	IV_{prov}
1995	e^{soe}	-12.42***	-15.36***	-13.63***	-14.44***
	$\ln FREV$	1.32***	0.88*	1.14**	0.73
First stage:	$\ln PROF^{soe}$	0.24*	0.24*	0.24*	0.19
	IV coefficient		0.69***	0.95***	0.89***
	st. error		(0.04)	(0.06)	(0.18)
	R ²		0.74	0.73	0.80
2004	e^{soe}	-8.61***	-12.30***	-14.44***	-18.75***
	$\ln FREV$	1.48***	1.08**	0.84*	0.80
First stage:	IV coefficient		0.62***	0.70***	0.85***
	R ²		0.54	0.48	0.64
2008	e^{soe}	-6.79***	-7.97***	-8.35***	-4.85
First stage:	IV coefficient		0.89***	0.70***	1.04**
	R ²		0.78	0.45	0.49

Note: *** – statistically significant at 1%; ** – at 5%; * – at 10%.

Entry Wedge and SOE Share, 1995-2004



SOE and NSOE Wages in s Prefectures

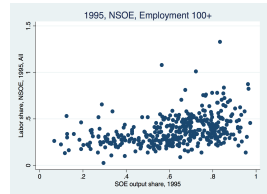
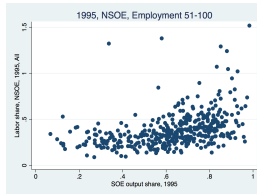
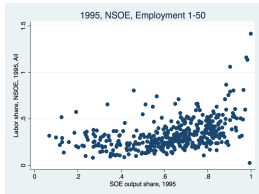


- SOEs pay the same wage in all s prefectures
- SOE and NSOE wages are similar in low s prefectures
- SOE wages are higher than NSOE wages in high s prefectures

Alternative Theory: Different Production Technologies

- NSOE firms in a prefecture have access to two technologies:
 1. inefficient low z technology with a high labor share (labor intensive)
 2. efficient high z technology with a low labor share
- A larger fraction of the NSOE firms in the high s prefectures will use technology 1 $\Rightarrow$ higher labor share
- Predictions of the alternative theory
 - within prefectures: smaller firms have higher labor share
 - across prefectures: conditional on size, firms have the same labor share

Alternative Theory: Different Production Technologies



- Predictions of the alternative theory are not consistent with the data
- Within prefectures
 - : firms with different sizes have the same labor share
- Across prefectures
 - : conditional on size, firms have increasing in s labor share

A politico-economic motivation for wedges

- Central government dictates a prefecture-specific target level of state employment, $\bar{N}_{SOE}$
- Problem: SOEs compete with private sector for workers
- Instruments: Local government use wedges $\{\tau^y, \tau^k, \psi\}$ to deliver $N_{SOE} = \bar{N}_{SOE}$
- Objective: Local government maximize entrepreneur profits conditional on z (want to “help a friend”)
- Optimal policy: set $\tau^y = \tau^k = 0$ and use ψ to constrain NSOE entry to ensure $N_{NSOE} = 1 - \bar{N}_{SOE}$

Introduce State-owned firms (SOE)

- Assume unit measure of potential SOE (and unit measure of potential NSOE)
- SOEs have same production function and same productivity distribution as NSOE
- SOEs compete with NSOEs for workers
- Key friction: central government decides what local state employment must be: $N_{SOE} = \bar{N}$
 - Local government must impose frictions on NSOE to satisfy employment constraint
- For simplicity: assume $\tau_y^{SOE} = \tau_k^{SOE} = 0$

Equilibrium in model with SOE and NSOE

- Labor market equilibrium requires $N_{NSOE} = 1 - \bar{N}$, implying

$$\frac{1 - \bar{N}}{\bar{N}} = (1 - \psi)(1 - \tau_y)^{\frac{\xi}{1-\eta}} \left(\frac{1}{1 + \tau_k} \right)^{\xi \frac{(1-\alpha)\eta}{1-\eta}}$$

- Note: target employment $\bar{N}$ is increasing in each of the wedges, (ψ, τ_k, τ_y)
 - an increase in $\bar{N}$ must be offset by an increase in ψ , τ_y , or τ_k (since ψ , τ_y , and τ_k are increasing in $\bar{N}$)

Equilibrium (cont.)

- Calculate profits – net of wedges – conditional on z and obtaining a licence;

$$\frac{\Pi(z)}{z} = \frac{1 - \bar{N}}{1 - \psi} \cdot (1 - \eta) \left(\frac{\xi z}{\xi - 1} \left(\frac{1 - \eta}{v} \right)^{\xi - 1} \left(\frac{(1 - \alpha) \eta}{r + \delta} \right)^{\xi \frac{(1 - \alpha) \eta}{1 - \eta}} \frac{1}{\bar{N}} \right)^{\frac{1 - \eta}{\xi \alpha \eta + 1 - \eta}}.$$

Politico-economic problem

- Local official choose wedges (ψ, τ_y, τ_k)
- Assume: official maximizes profits entrepreneur profits conditional on z , subject to
 1. a hiring constraint $N_{SOE} = \bar{N} \geq 1/2$
 2. wedges are non-negative, $\psi \geq 0$, $\tau_y \geq 0$, and $\tau_k \geq 0$

Motivation: give advantage to friends

- **RESULT:** The constrained optimal choice of wedges (ψ, τ_y, τ_k) imply $\tau_k = \tau_y = 0$ and $\psi > 0$.
- Expect to see a high correlation between SOE employment N_{SOE} and entry barrier ψ